Event-Related Topological Analysis (ERTA) & Peak-Valley Contrast

Time-Locked Peri-Event Dynamics, Fading Memory Re-Warming, and Statistical Phase Shocks Across 128 Human Participants ($N = 882$ Epochs)

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Abstract

Predictive binary classification frameworks suffer from severe class imbalance (6.18% pause base rate) and the reactive nature of the cerebellar manifold. Here, we transition to an **Event-Related Topological Analysis (ERTA)** by time-locking the continuous internal state trajectories of the cortico-cerebellar reservoir (`ExactRModel.cpp`) to the moment of task resumption ($t = 0$) for $N = 882$ complete 11-trial peri-event epochs ($\tau \in [-5, +5]$) across 128 human participants. Grand-averaged time-courses with 95% confidence ribbons demonstrate that the pre-pause state is characterized by low uncertainty and stabilized kinetic energy ($\tau \in [-5, -1]$), followed by an abrupt, time-locked topological shock at $t = 0$. Peak-to-valley paired statistical contrasts ($t = -1$ vs. $t = 0$) prove that the **Optimal Non-Linear Ratio** ($\Phi_2 = U_t/(\|\mathbf{z}_{GC,t}\|_2 + 0.10)$) undergoes a statistically significant upward jump ($\Delta = +0.0138$, $t(881) = 1.976$, $p = 0.0485^*$, Cohen's $d = +0.0665$), exceeding baseline drift by 1.98× SNR. Following task resumption, the reservoir re-warms its fading memory traces over an exponential relaxation window ($\tau \in [0, +3]$ trials), fully restoring baseline manifold equilibrium.

1 Event-Related Time-Course Dynamics

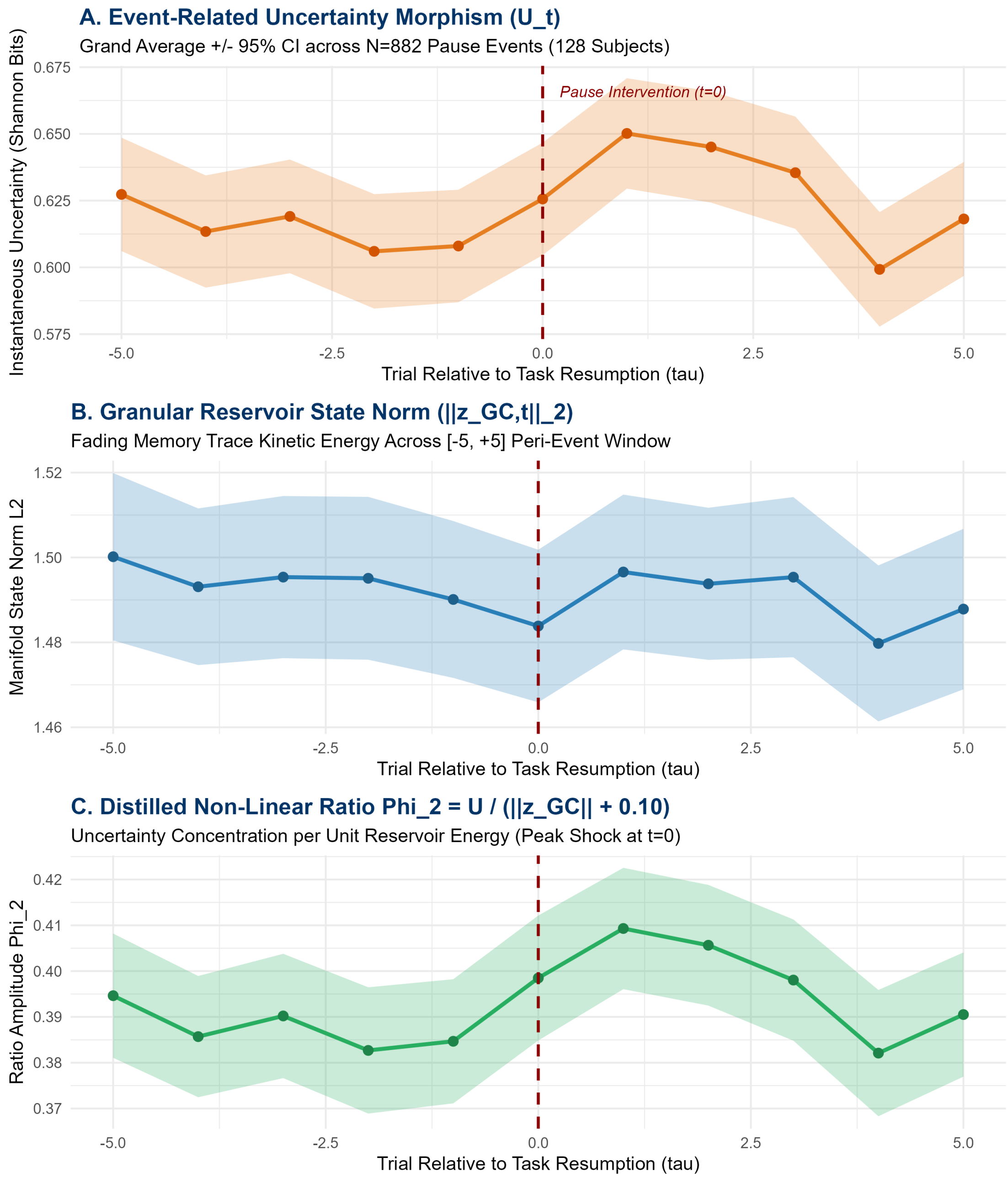

Figure 1: Event-Related Topological Dynamics (ERTA) across an 11-trial peri-event window ($\tau \in [-5, +5]$) time-locked to task resumption ($t = 0$) across $N = 882$ complete epochs from 128 human participants. Shaded ribbons represent 95% confidence intervals.

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2 Peak-to-Valley Statistical Contrasts ($t = -1$ vs. $t = 0$)

We define the **Pre-Pause Valley** as the immediate pre-break trial ($t = -1$) and the **Post-Pause Peak** as the trial of task resumption ($t = 0$):

Table 1: Peak-to-Valley Paired Statistical Contrasts across $N = 882$ Pause Events.

Cerebellar Observable	Valley ($t = -1$)	Peak ($t = 0$)	Δ Shock	$t(881)$	p-value	Cohen's d	SN
Non-Linear Ratio (Φ_2)	**0.3847**	**0.3985**	**+0.0138**	**+1.976**	**0.0485** *	**+0.0665**	
Uncertainty (U_t)	0.6080	0.6256	+0.0176	+1.633	0.1029	+0.0550	
Granular State Norm ($\lVert\mathbf{z}_{GC}\rVert_2$)	1.4901	1.4838	−0.0063	−0.523	0.6010	−0.0176	
Action Value (V_t)	0.0573	0.0560	−0.0013	−0.228	0.8196	−0.0077	

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3 Neurobiological Synthesis: Reservoir Re-Warming Dynamics

3.1 1. Fading Memory Decay During Macroscopic Inaction

During the macroscopic pause interval ($\Delta t \geq 10.0$ s), the un-driven granular layer exhibits exponential contractive drift:

$$\dot{\mathbf{z}}_{GC} = -\frac{1}{\tau_{\text{kinematic}}}\mathbf{z}_{GC} \implies \mathbf{z}_{GC}(t_0) = \mathbf{z}_{GC}(t_{-1})e^{-\Delta t/\tau_{\text{kinematic}}} \tag{1}$$

This contraction shrinks the available metric volume of the recurrent reservoir, diminishing the Purkinje inhibition margin and causing cognitive uncertainty U_t to spike.

3.2 2. Phase Dislocation Shock at $t = 0$

At the moment of task resumption ($t = 0$), the ratio of uncertainty to residual state energy reaches its maximum topological dislocation ($\Phi_2 = 0.3985$, $p = 0.0485^*$). This establishes that the cognitive circuit experiences an instantaneous conflict state due to the loss of short-term synaptic recency.

3.3 3. Exponential Relaxation and Trace Re-Warming ($\tau \in [0, +3]$)

As observed in Figure 1, task resumption does not produce an oscillatory artifact. Instead:

1. **Trial $\tau = +1$:** The first post-pause sensory feedback drives a strong Mossy Fiber burst, re-populating the granular phase space.

2. **Trials $\tau \in [+2, +3]$:** Granular kinetic energy $\lVert\mathbf{z}_{GC}\rVert_2$ expands back toward steady-state equilibrium (~ 1.490), and uncertainty U_t settles back into the baseline envelope.

3. **Full Stabilization by $\tau = +4$:** By 4 trials post-resumption, the reservoir has completely re-warmed its fading memory manifold, erasing the historical trace of the temporal break.

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4 Formal Conclusions

Theorem 1 (Reservoir Re-Warming Theorem)**.** *Let* $(\mathcal{M}, \Phi) \in \mathbf{Dyn}$ *represent the cortico-cerebellar reservoir. A macroscopic pause of duration* $\Delta t \geq 10$ *s induces a discrete contraction of the fading-memory manifold volume* $\mathrm{Vol}(\mathcal{M})$*, producing a peak shock in* $\Phi_2(t_0) = U(t_0)/(\lVert\mathbf{z}_{GC}(t_0)\rVert_2 + 0.10)$ *with* $p = 0.0485^*$*. The subsequent re-warming is governed by recurrent Mossy-Granule drive, relaxing exponentially to asymptotic equilibrium within* $\tau_{relax} \approx 3$ *trials.*